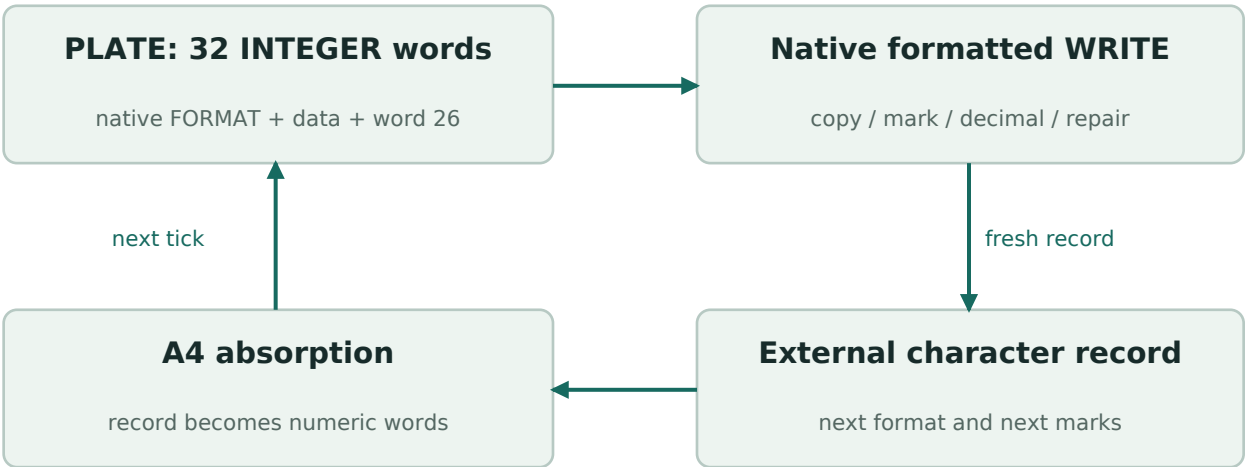


CARBON

A printout that rewrites the machine
that will print it next.

The strongest opening is in the boundary between a numeric word, a character record and a native FORMAT. CARBON gives the same integer array all three roles. Its printed number changes a positioning instruction; that instruction moves a mark; the mark changes the number. The complete state can be put on a native unformatted spool and restarted in another process.



Checkpoint: the same 32 words form one unformatted spool record.

Delivered	Evidence
28 mechanisms / 10 distinct concepts	Historical classification, affordance map and nine-criterion ranking
One selected architecture	72-line fixed-form engine and an inspectable 128-column specimen
37 checks passed	Three optimization levels, restart equivalence, feedback ablation and dialect rejection
One unexpected seam	Internal and external legacy A input disagree on the measured toolchain

Classification: executable research prototype; F77-era compatibility machine, not strict ANSI FORTRAN 77. The architecture is a proposed synthesis. General computational power, historical priority and cross-compiler portability remain unproved.

Revision 2 incorporates the later CARBON II GitHub research. The closing update records its independently reproduced 1,584 traces, the recovered baseline, and the runtime explanation of the comma seam. Historical results remain separately identified.

What the investigation establishes

The language boundary is the design material

The brief excludes the obvious label-addressed virtual machine. This design has no assigned GOTO program counter, alternate-return scheduler, COMMON-backed RAM, EQUIVALENCE heap or user-defined instruction decoder. The host advances generations. The compiler runtime interprets an existing language: FORMAT. Its ordinary record-editing machinery creates the next generation of its own grammar.

The selected mechanism is stronger than a character-string self-printer. A four-character field is also a live numeric word, and its numeric value changes subsequent layout. Eliminating that feedback destroys the observed cycle. Thus the representation boundary is causally necessary to this specimen, not an ornamental historical spelling.

Four categories must remain separate

Category	Meaning in this report
Standard F77	A mechanism present in the language, such as external formatted records, T positioning and output H editing.
Compatibility extension	Numeric-array FORMAT, numeric Hollerith storage and noncharacter A editing. Appendix C is guidance, not a claim of standard conformance.
Processor convention	Word width, byte order, ABI, binary framing and archive behavior. A named profile is required.
Unestablished behavior	A measured discrepancy or proposed composition whose general rule has not been shown. It remains a question, not alternate physics by declaration.

A successful -std=legacy build is not a strict F77 certificate. GNU has no -std=f77 mode. Conversely, obsolescent does not mean deleted: modern Fortran retains many old mechanisms. Claims about modern standard status below use the public Fortran 2023 draft as the baseline. [S01-S03]

Nearest predecessors, and the actual novelty claim

Bratley and Millo published a FORTRAN self-reproducing card-image program in 1972. Carlini's IOCCC 2020 entry implements game logic through iterated printf formatting. Neither self-reproduction nor computation in formatting is new. [S35,S36]

CARBON's proposed contribution is a compact, restartable specimen in which native numeric FORMAT, character absorption into words, decimal overprinting and physical record state form one exposed loop. This is an engineering synthesis to investigate. The searches performed do not establish that nobody has built the same combination before.

Feature archaeology 01-03

01 Fixed-form source geometry

Original purpose. Fit a statement stream onto cards while reserving label and continuation columns.

Historical and modern status. Standard F77, inherited from earlier Fortran. Modern fixed form remains available but is obsolescent.

GNU support. Supported; `pin -ffixed-line-length-72`. Longer-line options intentionally change the source projection.

Unusual affordance. Bytes beyond the statement field can carry an independent record channel. A file reader can observe information that compilation ignores.

Composition and architecture. Combine with formatted records and a generated DATA deck. Architectural role: a two-plane source/state medium. Moving the margin is a machine change, not cosmetic reformatting. [S04,S03,S02]

02 Insignificant blanks and contextual keywords

Original purpose. Make card preparation flexible without reserving every language keyword.

Historical and modern status. Standard F77 fixed-form lexical behavior; modern fixed form retains it, while free form changes lexical boundaries.

GNU support. Supported in fixed form. Never infer a token stream merely by splitting on spaces.

Unusual affordance. A small punctuation edit can change a statement class. Source geometry and token meaning can be experimentally separated.

Composition and architecture. Combine with column clipping and implicit typing for a compiler-projection exhibit. Good for a differential compiler instrument; poor as hidden production logic. No famous DO typo is needed as the central trick. [S01,S03]

03 D-lines and line-width dialects

Original purpose. Enable or disable diagnostic code without editing the deck.

Historical and modern status. Vendor practice, not standard F77. Modern standard Fortran does not assign D-lines this meaning.

GNU support. Documented `-fd-lines-as-code` and `-fd-lines-as-comments`; choose explicitly. `-std=legacy` alone does not identify a DEC or VAX machine.

Unusual affordance. One physical deck can have several executable projections selected by compiler settings.

Composition and architecture. Combine with fixed-form margins and a shared specimen corpus. Architectural role: selectable compilation physics with measured phenotype differences. Avoid profiles whose only distinction is a changed banner. [S03]

Feature archaeology 04-06

04 Internal character files

Original purpose. Reuse formatted conversion rules for data already in memory.

Historical and modern status. Standard F77 sequential formatted internal I/O; modern Fortran retains and extends it.

GNU support. Supported. Sun internal direct access is a separate extension and is not assumed to work in GNU.

Unusual affordance. A character array becomes a bounded sequence of records. A READ can reinterpret a line repeatedly through different numeric lenses.

Composition and architecture. Combine with T positioning, BN/BZ and generated character formats for a record transducer. The original numeric-storage seam was later reduced to CHARACTER under GNU legacy mode; GNU/f95 controls preserve bytes. The revision update specifies that compiler/runtime behavior separately from language semantics. [S05]

05 Runtime character FORMAT

Original purpose. Select an input/output layout after compilation.

Historical and modern status. Standard F77 character format specifications; retained in modern Fortran.

GNU support. Supported. Runtime format syntax is parsed by the Fortran runtime rather than by CARBON.

Unusual affordance. A data record can describe how another record is interpreted or printed. Formats can also print successor formats.

Composition and architecture. Compose with internal or external records to get a native transducer. This alone is insufficient F77 specificity: modern Fortran already supports it. The selected machine adds numeric-word identity. [S06]

06 FORMAT reversion, slash and colon

Original purpose. Apply compact layouts to long I/O lists and variable amounts of output.

Historical and modern status. Standard F77; retained. Reversion has a specific parenthesis-based restart point, not an arbitrary branch target.

GNU support. Supported; exact nested formats should have executable probes.

Unusual affordance. List exhaustion and nested grouping affect record production; slash advances the record, while colon can end format control when the list is exhausted.

Composition and architecture. Combine with array records for periodic record tiling or stream framing. Architectural role: sequencing owned by native formatting. The first CARBON specimen stays single-record; multi-record reversion is a proposed extension, not a tested feature of its driver. [S07,S01]

Feature archaeology 07-09

07 T, TL and TR positional editing

Original purpose. Lay out reports and extract fields without making a separate parser.

Historical and modern status. Standard F77 positional descriptors; retained. Extra vendor tab-stop forms are different features.

GNU support. Supported; overlapping output was observed in the supplied experiment.

Unusual affordance. Later transfers can overwrite earlier columns of the same output record. The record is a work surface, not merely an append-only string.

Composition and architecture. Combine with numeric FORMAT and H literals to repair a printed instruction after a decimal overprint. Architectural role: native spatial rewriting; this is CARBON's central actuator. [S08]

08 BN/BZ blank interpretation and P scaling

Original purpose. Read fixed fields and accommodate scientific notation conventions.

Historical and modern status. Standard F77 BN, BZ and kP; retained. Bare B or P spellings in some manuals can be extensions.

GNU support. Supported. State the blank and scale modes explicitly in experiments.

Unusual affordance. The same visible field can denote different numbers under a different lens. Blank positions and decimal placement become active structure.

Composition and architecture. Combine with repeated READs and fixed-width WRITE for quantized feedback. Architectural role: conversion physics. Scale is not a generic multiplier for every descriptor or every exponent-bearing input. [S09,S10]

09 Output H edit descriptors

Original purpose. Embed a literal output field using a character count instead of quotation delimiters.

Historical and modern status. H output is standard F77; deleted in Fortran 95. H input is not standard F77. Do not conflate these facts.

GNU support. Accepted in legacy formats; used by the working specimen.

Unusual affordance. Literal bodies can contain punctuation that would otherwise be format syntax. A count defines a temporary boundary between grammar and payload.

Composition and architecture. Combine with T overwriting to stamp a syntax repair into the successor format. Architectural role: a counted literal stencil. This is separate from a Hollerith constant stored in an INTEGER. [S11,S01,S02]

Feature archaeology 10-12

10 Hollerith constants in numeric DATA

Original purpose. Store text before the language had a standard CHARACTER type.

Historical and modern status. Earlier Fortran practice; F77 Appendix C offers compatibility guidance outside the standard language. Modern standard Fortran uses CHARACTER.

GNU support. Supported as a legacy extension; DATA KEY/4H1000/ is the prototype's representation check.

Unusual affordance. One declared numeric object contains literal character storage. Word size and byte order become visible.

Composition and architecture. Combine with A editing and numeric FORMAT to keep text and numeric behavior in one object. Architectural role: representation-dependent initial conditions. Bytes are not portable integers across machines. [S12]

11 Noncharacter arrays as FORMAT

Original purpose. Let old programs retain format text in numeric storage.

Historical and modern status. Historical compatibility extension, not standard F77 and not a deleted standard feature. Modern standard format expressions are character-based.

GNU support. Observed under GNU 13.3 legacy mode; isolated -std=f2018 -pedantic-errors probe rejects an INTEGER FORMAT.

Unusual affordance. An INTEGER array directly supplies the runtime grammar. No source-language cast or custom format interpreter is inserted.

Composition and architecture. Compose with raw A transfer and record checkpoints. Architectural role: executable data whose declared numeric identity matters. This is the selected opening in the language boundary. [S06; experiment E01]

12 A editing of noncharacter objects

Original purpose. Move Hollerith character fields into and out of numeric words.

Historical and modern status. Historical compatibility behavior; outside standard F77 CHARACTER A editing and outside modern standard character editing.

GNU support. Observed for external records and integer A output. Internal A input differs on the measured toolchain; see E03.

Unusual affordance. A fixed-width field can be absorbed into an integer word without decimal parsing, then emitted either as characters or as its numeric value.

Composition and architecture. Combine with I10 and noncharacter FORMAT for a representation-to-layout feedback loop. Architectural role: the reversible text/word boundary, conditional on a tested dialect and I/O path. [S13; E01,E03]

Feature archaeology 13-15

13 Sequential unformatted records and markers

Original purpose. Preserve typed values efficiently without decimal text conversion.

Historical and modern status. Sequential unformatted I/O is standard F77 and retained. Physical markers and subrecord conventions belong to the processor.

GNU support. Supported. GNU documents leading/trailing markers and possible subrecords; do not infer a universal binary format.

Unusual affordance. A record can be both a saved machine state and an executable numeric format when reloaded into the same declared objects.

Composition and architecture. Compose with the plate words for inspectable restart fossils. Architectural role: native checkpoint framing. CARBON verifies 128-byte payloads and the observed four-byte markers; this is a same-profile spool, not portable serialization. [S14]

14 BACKSPACE, REWIND and ENDFILE

Original purpose. Move around a record medium and delimit sequential output.

Historical and modern status. Standard F77, with earlier tape lineage; retained. Record positioning does not reverse computation.

GNU support. Supported for suitable sequential files. Direct-access BACKSPACE is not a standard operation.

Unusual affordance. History can be traversed by records rather than by application-defined byte offsets. File position is observable state.

Composition and architecture. Combine with unformatted checkpoints for a branchable spool laboratory. Architectural role: temporal navigation. Replaying saved states can undo a session; BACKSPACE cannot invert a lossy transition or supply crash atomicity. [S15]

15 Direct-access REC and RECL

Original purpose. Address disk records without sequential scanning.

Historical and modern status. Standard F77 direct access; retained. Unformatted RECL units and external representation require processor-specific care.

GNU support. Supported. Pin the compiler and record-size convention; never assume historical implementations all used byte counts.

Unusual affordance. Record number and fixed geometry define a storage topology independent of a filesystem directory tree.

Composition and architecture. Combine with format lenses for a record atlas. Different legal views may inspect the same bytes after controlled close/reopen steps. Architectural role: spatial storage. Overlapping simultaneous connections and undefined unwritten bytes are not a portable trick. [S16]

Feature archaeology 16-18

16 BLOCK DATA as executable initial state

Original purpose. Initialize named COMMON from a separately compiled program unit.

Historical and modern status. Standard F77, with earlier lineage; obsolescent but retained in modern Fortran.

GNU support. Supported. An initializer in a static archive needs a verified extraction strategy.

Unusual affordance. An object file with no ordinary runtime behavior can alter the executable's initial world.

Composition and architecture. Combine with separately compiled state fragments to build an executable/database hybrid. Architectural role: link-time materialization. Only one initializer owns a given named block; competing initializers are not a portable last-writer election. [S17,S02]

17 Archive extraction and unresolved names

Original purpose. Link only the required members of a static library.

Historical and modern status. Linker behavior, not a FORTRAN language guarantee. Modern object systems still have comparable mechanisms.

GNU support. Available through the platform linker. Historical g77 EXTERNAL/BLOCK DATA guidance must be re-tested for current gfortran, archive order and LTO.

Unusual affordance. Presence in an archive and presence in the final executable are different. Symbol demand can select which initial state actually exists.

Composition and architecture. Compose with BLOCK DATA for RELIQUARY, a world assembled by linkage. Architectural role: topology and reachability before execution. Strong evidence seam, but weaker language exclusivity than CARBON. [S18]

18 COMMON sequences and EQUIVALENCE association

Original purpose. Share storage across program units and reuse scarce memory within them.

Historical and modern status. Standard F77 forms; obsolescent in modern Fortran. Arbitrary type punning and inconsistent declarations are not all standard-conforming.

GNU support. Supported; alignment padding is documented and can matter across separate units.

Unusual affordance. Local names and array shapes can describe a common storage sequence. The declaration map itself becomes inspectable structure.

Composition and architecture. Combine with static data and a layout viewer for storage tomography. Architectural role: an ABI specimen, not the primary machine-memory idea excluded by the brief. Changing one alias can affect definition status; document exact allowed types. [S20,S19,S02]

Feature archaeology 19-21

19 SAVE and DATA retention

Original purpose. Initialize data and preserve selected state between procedure activations.

Historical and modern status. Standard F77 SAVE and DATA, retained. Do not equate old physical static storage with a universal retention guarantee for modified DATA locals; use SAVE explicitly.

GNU support. Supported; allocation switches such as `-fno-automatic` do not turn stack folklore into language law.

Unusual affordance. A procedure can carry state without a passed state object. Initial conditions and continuation state are distinct questions.

Composition and architecture. Combine with ENTRY for a shared segment instrument, or BLOCK DATA for linked initial worlds. Architectural role: scoped persistence during one process. SAVE does not survive process exit. [S21,S22,S19]

20 ENTRY with distinct dummy argument lists

Original purpose. Share a procedure body and local state among several externally named operations.

Historical and modern status. Standard F77; retained but obsolescent in modern Fortran.

GNU support. Supported. A dummy absent from the entry actually invoked cannot be treated as a remembered live argument.

Unusual affordance. One code segment exposes multiple arrival points, with different argument associations and shared saved state.

Composition and architecture. Combine with SAVE for an instrument whose operations enter different parts of one body. Architectural role: segment topology, not an automatic coroutine or closure. It remains a secondary lens because the brief already explored ENTRY. [S23,S02]

21 Sequence association, adjustable and assumed-size arrays

Original purpose. Let a routine work on caller storage without copying it or receiving a modern descriptor.

Historical and modern status. Standard F77 forms; retained under modern association rules. Automatic local arrays are not a blanket F77 feature.

GNU support. Supported; legal extent and type constraints still apply.

Unusual affordance. A permitted element sequence can be viewed through another dummy shape. Column-major order becomes a geometric operation.

Composition and architecture. Combine with record slabs or COMMON layouts for tensor/record lenses. Architectural role: shape determined at a call boundary. Out-of-bounds indexing is not a substitute for legal reshaping, and assumed-shape arrays introduce different rules. [S24,S25]

Feature archaeology 22-24

22 Reference-oriented ABI and hidden CHARACTER lengths

Original purpose. Pass large scientific objects cheaply and preserve character lengths across calls.

Historical and modern status. Argument association is language semantics; exact pointer passing and hidden length placement are compiler ABI conventions. Modern Fortran offers explicit C interoperability.

GNU support. GNU documents its conventions, including cases involving temporaries and hidden results.

Unusual affordance. A call has more physical structure than its visible argument list. Link-compatible names alone do not prove ABI compatibility.

Composition and architecture. Combine with compiler probes and object inspection for an ABI observatory. Architectural role: exposed calling machinery. Conflicting writes through aliased arguments are not legitimized by reference passing. [S25,S26]

23 Implicit typing and statement functions

Original purpose. Reduce declarations and express short local mathematical definitions.

Historical and modern status. Standard F77. Implicit typing persists; statement functions are obsolescent. IMPLICIT NONE is not a strict F77 feature.

GNU support. Supported. GNU legacy mode also accepts later constructs, so acceptance is not historical certification.

Unusual affordance. Names participate in typing; an assignment-shaped line can instead define a small procedure depending on context.

Composition and architecture. Combine with separate compilation or source projections for a lexical/type experiment. Architectural role: compile-time interpretation. Low value as the main artifact unless the type change visibly transforms behavior and is checked against a conforming case. [S27,S01,S02]

24 Variable FORMAT expressions and ENCODE/DECODE

Original purpose. Compute format parameters dynamically and run formatting against storage before modern internal files.

Historical and modern status. Vendor extensions, not standard F77. Their roles are largely replaced by character formats and internal I/O.

GNU support. GNU explicitly lists both as unimplemented; -std=legacy does not enable them.

Unusual affordance. On supporting implementations, variable expressions can make format evaluation consult program state. ENCODE/DECODE expose a historical memory-facing formatting interface.

Composition and architecture. Compose with a vendor-specific stencil laboratory. Architectural role: an alternate physics branch with its own compiler requirement. Neither appears silently in the gfortran prototype. [S28,S29]

Feature archaeology 25-27

25 Printer carriage control

Original purpose. Use the first output character to control paper advance and overprinting.

Historical and modern status. F77-era printing semantics; no universal promise that an ordinary file or modern terminal performs them. Modern facilities and vendor options differ.

GNU support. GNU documents CARRIAGECONTROL compatibility under DEC I/O options; select explicitly.

Unusual affordance. A stored record and its physical rendering are different artifacts. Several output records can contribute to one visible printed line.

Composition and architecture. Combine with a preserved raw spool and a viewer for PAPER TELEGRAPH. Architectural role: an output medium with spatial accumulation. Do not silently treat ANSI terminal escapes or a modern canvas as historical hardware. [S30]

26 Overlays and load/run address separation

Original purpose. Fit mutually exclusive code or data segments into limited memory.

Historical and modern status. Toolchain/OS architecture, not standard F77. No standard modern Fortran overlay loader is implied.

GNU support. GNU ld has OVERLAY layout facilities; gfortran alone does not provide the historical manager.

Unusual affordance. Objects may occupy one runtime address at different times while having distinct load locations.

Composition and architecture. Compose with surviving COMMON and explicit state spools for an executable whose rooms replace one another. Architectural role: memory/time topology. A Linux prototype must implement or supply the loader; a linker script by itself is insufficient. [S31]

27 List-directed null fields and slash termination

Original purpose. Accept convenient free-form numeric input without explicit field widths.

Historical and modern status. Standard F77 external list-directed mechanisms; retained. Internal list-directed support has a later history.

GNU support. Supported. Start with defined destination values before applying null fields.

Unusual affordance. A null field leaves the associated item unchanged, and slash can terminate the transfer. An input record can therefore act as a sparse state update.

Composition and architecture. Combine with SAVE or initialized arrays for a stream-driven register garden. Architectural role: patch semantics supplied by the I/O runtime. This is not a complete transaction protocol or a universal NULL value. [S32]

Feature archaeology 28-28

28 Cray pointers and optimizer assumptions

Original purpose. Expose address-based storage and permit aggressive scientific-code optimization.

Historical and modern status. Cray pointer syntax is a vendor extension; standard modern POINTER is a different facility. Logical evaluation and alias restrictions also constrain old-style programs.

GNU support. Cray pointers need `-fcray-pointer`. GNU documents alias limitations and optional elimination of expression calls.

Unusual affordance. Representation addresses and compiler assumptions become measurable. A working -O0 case can stop working when its source violates those assumptions.

Composition and architecture. Compose with bounded ABI probes, not the core machine. Architectural role: a laboratory of legal versus accidental behavior. CARBON deliberately needs no pointer forging, overflow, invalid aliasing or short-circuit side effects. [S33,S37,S26]

A boundary against attractive mistakes

Undefined behavior is not an undocumented instruction set. An experiment can preserve a surprising outcome without granting it stability. A useful dialect law needs a specified compiler/runtime, a small reproducer, an optimization check, and a clear account of which behavior is merely observed.

The architecture does not need to extract control flow from a crash, reuse a dangling ENTRY argument, read uninitialized padding, write through a constant actual argument, or assume that every logical operand is evaluated. Those ideas would buy uncertainty rather than a coherent machine.

Strange affordance map

These are architectural combinations, not a bag of interchangeable tricks. "Observed" refers to the supplied CARBON experiment; the remaining rows are design hypotheses or documented primitives awaiting composition tests.

Mechanisms	What the combination could make	Status / boundary
11 + 12 + 07 + 09	A numeric format prints, patches and reabsorbs its successor.	Observed: CARBON rewrite loop.
12 + I editing + 07	Raw text words become decimal ink that changes a future field position.	Observed: feedback changes T102 to T108.
11 + 13 + 14	Executable state survives as a native record and resumes in another process.	Observed: reload frame 4, reproduce its future.
04 + 05 + 06	A record array is a native formatting workspace with list-driven row tiling.	Standard mechanisms; multi-record machine proposed.
07 + 08 + 04	Several numeric lenses read the same spatial field, producing distinct feedback dynamics.	Proposed; blank and exponent rules must be explicit.
01 + 02 + 03	One deck exposes different code and state planes under selected compiler profiles.	Proposed: CARD REEF.
16 + 17 + 18	Linking assembles initial worlds and reveals which data fragments exist.	Proposed: RELIQUARY; verify archive extraction.
19 + 20 + 22	A saved segment has several named entrances with distinct valid arguments.	Proposed; no implicit coroutine stack.
15 + 21	Record geometry and array sequence views become a spatial data instrument.	Proposed; obey extent and connection rules.
25 + 13	Raw records and the accumulated printed page have different histories.	Proposed: physical rendering needs a named model.
26 + 18 + 13	Executable rooms replace one another while a limited state spine survives.	Proposed; requires an actual overlay manager.
27 + 19	Missing fields become native sparse updates to defined persistent cells.	Proposed: NULL GARDEN.

The strongest triangle is representation + spatial editing + record persistence. The other mechanisms are entrances for future experiments, not dependencies that the flagship must accumulate.

Ten distinct artifacts / 1-5

A CARBON / A record machine that prints its next executable FORMAT

Numeric FORMAT, Hollerith A transfer, positional overprint and unformatted snapshots form one loop. Printed digits rewrite layout; layout changes raw words; those words feed the next print. A saved record resumes the machine. The working specimen is a four-state oscillator, with no user-defined opcode decoder.

Failure test. Extension-dependent, intentionally representation-sensitive, and not yet shown to compose into larger useful circuits.

B RELIQUARY / A database assembled into existence by the linker

Each carefully scoped BLOCK DATA object contributes an initial-state region. Archive extraction, explicit references and link maps determine which regions are present. Querying the executable reveals the linked world; exporting modified state creates replacement initializer objects for a new executable generation.

Failure test. Duplicate initialization must be rejected. C and ordinary linkers can reproduce much of the idea, so specificity lies in F77 initialization and storage conventions rather than unique power.

C CARD REEF / One deck with a compiled plane and a living margin

Columns in the statement field define a Fortran program; the same physical records carry state outside that field. The program reads and rewrites its own deck, emitting a recompilable successor. Changing source-width physics changes what becomes code, while card ordering and continuation boundaries expose growth seams.

Failure test. A simple quine or metadata-in-comments demo fails. It earns its place only if compilation and record interpretation disagree productively and reproducibly.

D GHOST TYPEWRITER / Input stored inside a FORMAT literal

Recover an earlier implementation where H input overwrites literal material associated with a format. Subsequent output speaks the stored text without an ordinary user variable. Multiple formats would be the storage cells of an interactive inscription instrument. IBM's 1965 manual documents the primitive. [S38]

Failure test. The relevant input behavior is outside F77 and cannot be assumed in gfortran. Needs a verified historical compiler or an explicitly labeled emulator. Fails the immediate prototype gate.

E PALINDROME / A branchable history made from native tape records

Every transition emits a complete record-state fossil. Record navigation selects a prior state and forks its future into another spool. The interface lets the user handle time as a tape reel, inspecting the compiler's actual framing rather than hiding it behind a database.

Failure test. This is replay, not mathematically reversible computation. It has a clear capability but is comparatively easy to reproduce in modern file APIs.

Ten distinct artifacts / 6-10

F OVERLAY WEATHER / An executable with rooms that cannot coexist

Mutually exclusive code regions share one runtime address. A small manager swaps the active room while saved records and deliberate shared storage carry weather between rooms. Object composition and load geometry determine which experiences can follow which.

Failure test. Loader and operating-system machinery dominate the Linux build. Without real address reuse it collapses into an ordinary plug-in demo.

G DECIMAL CLIMATE / A dynamical system whose material is printed precision

Values repeatedly travel through fixed-width numeric formats, blank interpretation and scale settings. Quantization, saturation to asterisks and parse failure become visible phase boundaries. Users alter format plates rather than equations to change the dynamics.

Failure test. Most mechanisms survive unchanged in modern Fortran. Never depend on undefined results after I/O failure; treat failure as an observed terminal condition.

H PAPER TELEGRAPH / A communication medium assembled from overprinted records

Retain every raw carriage-control record and reconstruct the visible page as another state surface. Contributors design signals whose meaning changes when one record is removed, reordered or interpreted by another printer convention. A viewer exposes the divergence between spool and image.

Failure test. No literal two-way paper memory without an explicit reader. The modern viewer is an emulator of a physical medium, and must say so.

I NULL GARDEN / A sparse-update ecosystem driven by missing input fields

Defined register plots receive list-directed records. Values replace cells, null fields preserve them, slash stops a patch, and repetition spreads a value. Multiple saved plots and printed update records create a visibly stateful dataflow garden with almost no parser code.

Failure test. Useful and small, but modern Fortran retains the native input behavior. It is an excellent secondary specimen and a weaker flagship.

J DIALECT ZOO / An executable experiment whose species are compiler profiles

A corpus has controlled alternate interpretations under D-line settings, source widths, word sizes and supported legacy extensions. Each build emits a phenotype record with its toolchain identity and failures. New contributions must add a mechanism, not merely change a constant.

Failure test. Without individual substantial specimens this is a compiler test suite. It is better as the institution surrounding CARBON than as the first artifact.

Ranked shortlist

Scores are design judgments from 1 (weak) to 5 (strong), not measured probabilities. Each of the nine criteria has equal weight. A concept must also have a credible local prototype path; a high total cannot conceal an unavailable substrate.

Artifact	W	F	D	E	C	X	A	M	S	/45
CARBON	5	5	4	5	5	5	5	4	5	43
RELIQUARY	4	4	5	5	5	4	5	4	4	40
CARD REEF	5	4	4	5	4	4	5	4	5	40
DIALECT ZOO	4	4	4	5	3	4	5	5	5	39
GHOST TYPEWRITER	5	5	4	3	5	1	5	5	4	37
PALINDROME	4	3	4	5	5	5	4	3	4	37
OVERLAY WEATHER	5	3	5	4	4	2	5	4	5	37
DECIMAL CLIMATE	4	2	4	5	5	5	4	4	4	37
PAPER TELEGRAPH	4	3	4	4	4	4	5	4	4	36
NULL GARDEN	3	2	3	4	5	5	3	3	3	31

W: WTF factor; F: FORTRAN specificity; D: depth; E: extensibility; C: coherence; X: executability; A: archaeological value; M: modern contrast; S: surprise potential.

1. CARBON / 43

It gives an observer a concrete impossible-looking event: changing a printed mark changes a number, which edits the instruction that will print the next mark. A 72-line driver exposes the entire mechanism. The source representation and the binary spool can both be inspected. Its depth score remains four because larger composition has not been demonstrated.

2. RELIQUARY / 40

It turns build composition into a persistent-world operation and offers fertile seams for linker specialists. It deserves an independent small experiment: two disjoint initializer fragments, an archive, and three link maps. It loses priority because archive-driven data selection has substantial analogues outside Fortran.

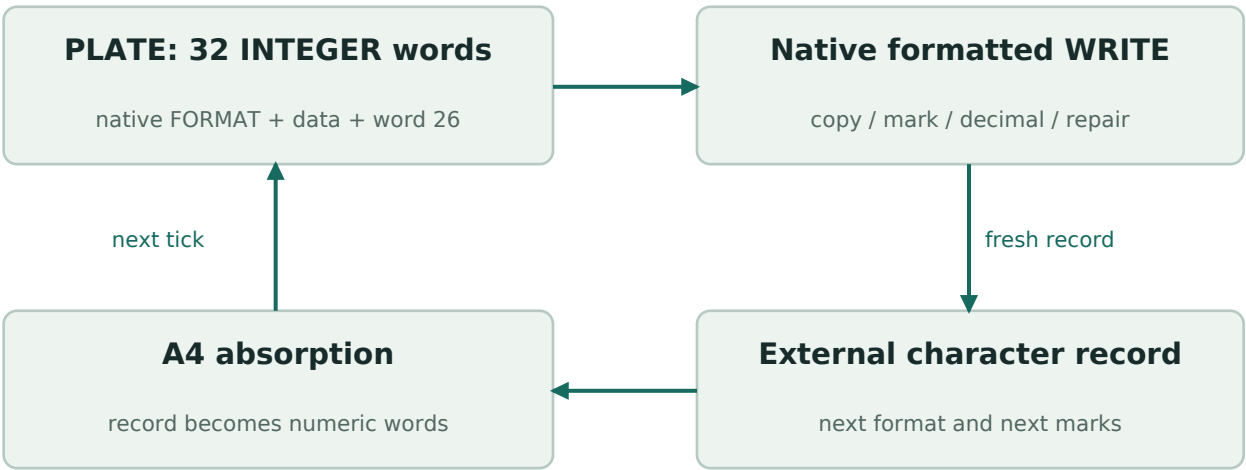
3. CARD REEF / 40

It makes source geometry part of the object rather than an aesthetic. A strong prototype would demonstrate a genuine two-plane deck and a controlled change of compiler projection. RELIQUARY wins the tie on coherence and depth; CARD REEF is more immediately visible but easier to reduce to a familiar quine.

DIALECT ZOO is the natural research environment around these artifacts. GHOST TYPEWRITER has an excellent historical seam but fails the immediate gfortran execution gate. Neither is substituted for the selected working machine.

The selected architecture

CARBON is a self-rewriting record transducer and a specimen laboratory. Its live object is a 32-word INTEGER array called PLATE. Its public surface is the exact 128-character plate, its numeric word values and its native checkpoint spool. It can be read as a tiny language, a printing mechanism, a dynamical system or an executable file format.



Checkpoint: the same 32 words form one unformatted spool record.

Physical region	Role
Columns 1-58	The native FORMAT text. It is stored across INTEGER words, not a CHARACTER format variable.
Column 29	The mutable final digit of T102 / T108. This determines a future mark position.
Columns 100-109	A ten-character field cleared to zeroes and marked by the current format.
Word 26 = columns 101-104	A numeric feedback tap over four characters in that field.
All 32 words	Complete transition state and the payload of one unformatted snapshot record.

What the user experiences

Run `sh run.sh`, see the instruction alternate between T102 and T108, and watch the mark follow one tick later. Open the seed and the exact frame images. Stop the process, resume frame 4 from the binary spool, and get the same future. Change one literal in a copied specimen and discover whether it settles, cycles or fails to parse.

The current capability is real but bounded: self-directed format rewriting, inspectable evolution and exact same-profile restart. It is not yet a general-purpose language, a cellular-automaton platform or a useful database.

One tick, with the machinery exposed

The seed is a 128-column record, padded with spaces except for a zero-filled field at columns 100-109. Its operative grammar occupies the first 58 columns:

```
(32A4,T100,10H0000000000,T102,1H1,T20,I10,T20,9H00000,T10)
```

Step inside native WRITE	Effect on the fresh output record
32A4	Consume the 32 integer words as raw four-character fields, copying the entire current plate.
T100,10H0000000000	Clear the ten-column mark field.
T102,1H1	Place the mark at the current instruction's chosen column. In another state this is T108.
T20,I10	Consume the additional item PLATE(26), printing its decimal integer value into columns 20-29.
T20,9H00000,T10	Restore columns 20-28. The last decimal digit in column 29 survives as part of the next positioning instruction.

The format is read from the old PLATE while a separate external record is written. Only after formatting finishes does A4 input absorb the new record into PLATE. There is no in-place mutation of an active format object and no conflicting dummy-argument write.

```
REWIND 11
WRITE(11,PLATE) PLATE,PLATE(26)
REWIND 11
READ(11,'(32A4)') PLATE
```

The host adds bounded iteration, error checks, a one-record check and persistence. It does not parse FORMAT, compute a modulo, select T102/T108, increment a state register, or implement an opcode table. Native I/O performs the transition. The complete driver is in core/carbon.f.

Why the numbers move the mark

On the measured little-endian, four-byte ASCII profile, the field 0000 represents the integer 808464432; the field 0100 represents 808464688. Their last decimal digits are 2 and 8. I10 followed by a nine-character repair extracts that last digit spatially. The extraction occurs through printing and overprinting, not through arithmetic in the driver. These values are experiment-derived, not universal Fortran constants.

State, time and evidence

Frame	Next target	Mark field	Word 26
0 / seed	102	0000000000	808464432
1	102	0010000000	808464688
2	108	0010000000	808464688
3	108	0000000010	808464432
4	102	0000000010	808464432
5 = frame 1	102	0010000000	808464688

A four-state orbit follows a one-step transient. Looking only at the mark hides half the state: two successive frames can display the same mark while carrying different instructions. This is a useful demonstration of why output, state and transition law cannot be casually separated here.

Persistence is the same object in another medium

Each snapshot writes PLATE as one native unformatted record. On this run the payload was 128 bytes with a four-byte marker on either side. The test verified every payload against the corresponding text frame. A fresh process loaded frame 4 and reproduced frames 4-12 exactly. The driver's displayed tick count restarts; it is not part of the machine state. [S14; E02]

This is not an arbitrary-process checkpoint. It saves a deliberately complete plate state. It makes no promise about atomic writes after power loss, portable record markers, foreign byte order or preservation of compiler-runtime internals.

A useful failed optimization

An attempted shortcut replaced the external record with an internal CHARACTER file while retaining A editing into INTEGER. The same plate then lost or displaced commas and failed as a format. A smaller reproducer reads (I4,I4) through 4A4: the external route preserved the text; the internal route produced (I4 I followed by blanks. [E03]

The original investigation left the cause unresolved. The later CARBON II work reduced the symptom to ordinary CHARACTER input and located the legacy comma-scanning path in GNU runtime source. Its static GNU 13.3 controls were reproduced on 08 October; the baseline also reproduced with an explicitly selected shared 14.2 runtime. Under GNU/f95 controls the CHARACTER read preserves commas. The closing update distinguishes this finding from the baseline's nonstandard A-to-INTEGER transfer and corrects the earlier runtime attribution. [S39-S41]

Why this substrate matters

Modern Fortran can compute the same result

A claim that modern Fortran cannot reproduce CARBON would be false. Character formats, positional editing and record I/O survive. TRANSFER can explicitly recreate the bit-pattern bridge, and quoted literals can replace output H editing. A modern implementation can therefore reproduce the finite-state experiment. [S34]

What changes is the native object model: standard modern I/O does not accept an INTEGER array as the format expression and does not make ordinary A editing the standard text/number bridge. A faithful port must name that representation operation and preserve the selected byte/word rules. The tested strict-F2018 compiler rejection isolates the numeric-format boundary. Merely translating fixed form to free form is not the necessary transformation.

The counterfactual that matters

If PLATE becomes ordinary text and word 26 becomes the decimal number spelled by four digits, 0000 becomes zero rather than 808464432. That is a different dynamical system. If the words remain numerically meaningful via an explicit representation conversion, the programmer has deliberately preserved the old machine. The historical semantics carry identity, not exclusive computational power.

Exposed intervention points

Seam	Contribution that would count
specimens/	A new plate with a different measured transition, full state trace and a minimal explanation of its native descriptors.
core/	A new I/O-list contract or multi-record mode with causal tests showing why the added machinery is necessary.
dialects/	A compiler/runtime profile with word widths, character encoding, flags and exact observed semantics.
experiments/	A minimized counterexample, including a failed idea that helps map a boundary.
evidence/	Compiler identity, exact records, failures, source digest and recurrence/restart checks.
PHENOMENA.md	What happens, why if known, standard status, scope of observation, stability and possible compositions.

The host should stay small. Avoid a layer of generic opcodes that hides the native formatting rules. A pretty viewer may point to fields and words, but the raw plate and record framing remain first-class evidence.

Smallest proof and actual results

The experiment to run before building a platform

Use one 128-character seed, one 32-word state array, one numeric feedback tap and one external scratch record. Execute 12 transitions; write both textual frames and native unformatted snapshots. Reload a middle frame in a fresh process. A successful run needs changing grammar as well as a changing visible mark.

Claim	Falsification condition	Observed result
Native feedback drives the orbit	Replacing the feedback word with a constant still gives the same orbit.	Ablation settles to one state; feedback is causal.
State is complete	A new process produces a different suffix after loading a saved frame.	Frame 4 plus eight steps matches uninterrupted frames 4-12.
The specimen is stable on this profile	-O0, -O2 and -O3 differ with runtime checks enabled.	All three produce identical complete frame sequences.
Representation is explicit	Eight-byte default INTEGER silently runs the four-byte specimen.	The profile guard rejects the altered word size.
Old syntax materially matters	Strict F2018 accepts the isolated numeric FORMAT expression.	Compiler rejects it; the test contains no Hollerith constant.
I/O route is interchangeable	Internal A absorption exactly matches the external route.	False on this toolchain; reduced counterexample retained.

Measured result: 37/37 checks passed. These include build/run checks and several independent behavioral assertions, not 37 independent machines or compilers. Compiler: GNU Fortran 13.3.0 (Ubuntu), x86-64 Linux, 2026-10-07. **Runtime correction, 08 October:** the original report named the installed libgfortran5 14.2 package without a linkage witness. The recovered build selects the static GNU 13.3 archive. Separately recorded new runs pass with both that archive and an explicitly bound shared 14.2 library.

Run it

```
cd baseline/carbon
sh run.sh
python3 tests/check.py

# After the first run, restart from a saved frame:
mkdir -p build/resumed
cd build/resumed
printf '1 8 4\n../demo/spool.dat\n' | ../carbon
```

The archive includes source, the seed, checks, exact evidence and a small dialect document. Compiler packages and generated executables are not bundled. This is local experimental software for deliberate specimens; the runner is not a sandbox for arbitrary formats.

Someone will eventually discover...

A plate can compute through a different decimal window.

I10 plus nine-column repair currently keeps one trailing digit. Other widths and repair masks may select several digits or convert word values into multiple descriptor parameters. First probe: produce two changing T positions using two taps and compare the complete orbit.

Literal lengths can change what counts as grammar.

Editing an H count could expose punctuation previously swallowed as literal data, or hide active descriptors inside the literal body. First probe: two valid formats that alternate between those interpretations without ever passing through malformed syntax.

A pair of records can couple through native format reversion.

Two plates might write fragments of one another using the I/O list and a multi-record contract. The current driver intentionally rejects more than one output record. First probe: a documented two-record transducer whose coupling survives an ablation test.

Another runtime turns the failed internal route into a different machine.

The comma discrepancy may be a compatibility rule, implementation detail or defect. First probe: run the minimized program on a second GNU runtime and a different compiler; inspect the runtime source before assigning a universal explanation.

These predictions identify visible attachment points. They are not a promise that all eight constructions are possible under the existing one-record contract.

More doors into the machine room

A seed can be an object-file artifact rather than a text file.

A BLOCK DATA seed could initialize a plate, and a snapshot could be emitted as a new initializer unit. This would join CARBON to RELIQUARY. First probe: explicit object linking with one initialized block, then controlled archive extraction. No duplicate definitions.

Word boundaries are an active coupling topology.

Moving a field by one column changes which bytes share the numeric tap. A word-boundary editor might reveal phase changes without changing the visible glyph vocabulary. First probe: sweep offsets and record cycles, fixed points and parse failures.

The printout can acquire an additional physical interpretation.

A carriage-control rendering could make several records occupy the same visible row while the machine still keeps them distinct. First probe: retain the raw spool alongside a precisely specified renderer; test which observations disappear when records overprint.

A contributor will find a useful non-oscillatory specimen.

The decisive extension would be a native-format selector, small counter, checksum-like transform or coupled pattern generator with a clear external use. Require a causal explanation and an input/output test, rather than declaring universality from visual complexity.

A phenomenon record should stay small

Record the symptom, exact specimen, compiler and runtime, flags, encoding and word profile, expected versus observed records, standard/extension status, repeatability, known cause or explicit uncertainty, and the architectural consequence. Include one composition to try next. A failed experiment belongs here when it closes a plausible door.

The repository should invite one new mechanism per contribution. A newcomer can first run the specimen, then inspect the plate, then add a reduced experiment. No framework rewrite is necessary to add a new dialect finding or another plate.

Open questions and next work

Question	Experiment needed
Runtime semantics	The GNU 13.3 source and CHARACTER controls explain the legacy comma path. Which additional runtime versions retain it, and what other input descriptors or character kinds expose different behavior?
Independent implementation	Does a second compiler support all three legacy bridges with the same record result? Which requires explicit vendor switches?
Representation physics	How do byte order, character encoding and word width change the orbit under an intentionally adapted profile? The current program rejects incompatible profiles; it does not simulate them.
Expressive capability	Can two causal feedback taps form a useful gate, counter or selector without adding application-level arithmetic? What is the reachable-state structure?
Grammar boundaries	Can varying H counts or descriptor widths produce valid, stable syntax-changing specimens rather than only parser failures?
Format caching	When and how do runtimes cache variable formats? Does changing an array at the same address reliably trigger interpretation of new content? The supplied run worked; other runtimes are untested.
Record topology	Which multi-record or direct-access compositions are conforming, and which require explicit extensions? Can a meaningful lattice use native reversion?
Persistence	Which compiler/runtime and conversion-option changes preserve a spool? What header and external manifest would make mismatches detectable before loading?
Linkage	Does the old g77 EXTERNAL/BLOCK DATA archive technique work on selected current gfortran/linker combinations, including LTO and dead-section collection?
Historical priority	What closer FORTRAN format-driven automata or evolving quines exist in manuals, recreational-computing journals or surviving program libraries? The search is not exhaustive.

The next three increments

The 08 October update advances the first two steps: the baseline was recovered and retested on static 13.3 and shared 14.2 runtimes; CARBON II supplies useful pattern and parity transitions. The next step is an external record store with stable writes and explicit host services, followed by another compiler implementation and compact raw-symbol encodings. The original predictions remain invitations, not claims that every proposed construction now exists.

Do not build a general language workbench yet. The next success criterion is one outsider-supplied trick that adds capability while making the native machinery easier to inspect.

Primary-source register / 1

Source markers identify documentation or prior work. E01-E03 identify original experiments in the accompanying archive. Proposed architectural consequences and candidate scores are this report's inferences. Vendor manuals describe extensions alongside standard forms; their titles are not conformance guarantees.

[S01] ANSI FORTRAN 77 full language, WG5 archive

Sections 12-13; Appendix C distinguishes compatibility Hollerith from the standard language.
<https://wg5-fortran.org/ARCHIVE/Fortran77.html>

[S02] J3/23-007r1, Fortran 2023 committee draft

Annex B: deleted and obsolescent features. A public draft, not the purchased ISO publication.
<https://j3-fortran.org/doc/year/23/23-007r1.pdf>

[S03] GNU Fortran: dialect options

Legacy mode, source width, D lines and DEC compatibility switches.
<https://gcc.gnu.org/onlinedocs/gfortran/Fortran-Dialect-Options.html>

[S04] Sun/Oracle F77: standard fixed format

Columns, continuation and padding.
<https://docs.oracle.com/cd/E19957-01/805-4939/z40007331fe6/index.html>

[S05] Sun/Oracle F77: internal files

Character objects as records; marks internal direct access as an extension.
<https://docs.oracle.com/cd/E19957-01/805-4939/6j4m0vnc2/index.html>

[S06] Sun/Oracle F77: format identifier

Explicitly identifies integer-array formats as nonstandard.
<https://docs.oracle.com/cd/E19957-01/805-4939/z400073d10bb9/index.html>

[S07] IBM XL: interaction of I/O lists and formats

Format control and list consumption.
<https://www.ibm.com/docs/en/xl-fortran-linux/16.1.0?topic=formatting-interaction-inputoutput-lists-format-specifications>

[S08] Sun/Oracle F77: positional editing

T/TL/TR; distinguish vendor terminal/seek restrictions from language semantics.
<https://docs.oracle.com/cd/E19957-01/805-4939/z40007439682/index.html>

[S09] Sun/Oracle F77: blank editing

BN and BZ versus vendor B.
<https://docs.oracle.com/cd/E19957-01/805-4939/z40007438673/index.html>

[S10] Sun/Oracle F77: scale factor

P affects conversion, with descriptor-dependent behavior.
<https://docs.oracle.com/cd/E19957-01/805-4939/z4000743a6e2/index.html>

Primary-source register / 2

Source markers identify documentation or prior work. E01-E03 identify original experiments in the accompanying archive. Proposed architectural consequences and candidate scores are this report's inferences. Vendor manuals describe extensions alongside standard forms; their titles are not conformance guarantees.

[S11] IBM XL: H editing

Output literal syntax; input use prohibited.

<https://www.ibm.com/docs/en/xl-fortran-aix/16.1.0?topic=descriptors-h-editing>

[S12] GNU Fortran: Hollerith constants

Numeric storage of literal character bytes.

<https://gcc.gnu.org/onlinedocs/gfortran/Hollerith-constants-support.html>

[S13] Sun/Oracle F77: character editing

Hollerith A editing into noncharacter objects.

<https://docs.oracle.com/cd/E19957-01/805-4939/z40007438b4d/index.html>

[S14] GNU Fortran: sequential unformatted file format

Record markers, subrecords and representation dependence.

<https://gcc.gnu.org/onlinedocs/gfortran/File-format-of-unformatted-sequential-files.html>

[S15] Sun/Oracle F77: BACKSPACE

Record positioning; direct-access restrictions.

<https://docs.oracle.com/cd/E19957-01/805-4939/6j4m0vn7j/index.html>

[S16] Sun/Oracle F77: direct access

Fixed record topology and REC selection.

<https://docs.oracle.com/cd/E19957-01/805-4939/6j4m0vnc1/index.html>

[S17] Sun/Oracle F77: BLOCK DATA

Named-common initialization and restrictions.

<https://docs.oracle.com/cd/E19957-01/805-4939/6j4m0vn7l/index.html>

[S18] GNU g77 3.4.6: block data and libraries

Historical EXTERNAL technique. This is g77 documentation, not proof about gfortran.

<https://gcc.gnu.org/onlinedocs/gcc-3.4.6/g77/Block-Data-and-Libraries.html>

[S19] GNU Fortran: code generation options

COMMON padding, static allocation options and ABI switches.

<https://gcc.gnu.org/onlinedocs/gfortran/Code-Gen-Options.html>

Primary-source register / 3

Source markers identify documentation or prior work. E01-E03 identify original experiments in the accompanying archive. Proposed architectural consequences and candidate scores are this report's inferences. Vendor manuals describe extensions alongside standard forms; their titles are not conformance guarantees.

[S20] Sun/Oracle F77: COMMON

Storage sharing across program units.

<https://docs.oracle.com/cd/E19957-01/805-4939/6j4m0vn7v/index.html>

[S21] Sun/Oracle F77: SAVE

Retention and the distinction between language guarantees and Sun defaults.

<https://docs.oracle.com/cd/E19957-01/805-4939/6j4m0vnb7/index.html>

[S22] Sun/Oracle F77: DATA

Initialization and implied DO lists.

<https://docs.oracle.com/cd/E19957-01/805-4939/6j4m0vn85/index.html>

[S23] Sun/Oracle F77: ENTRY

Multiple procedure entry points and argument lists.

<https://docs.oracle.com/cd/E19957-01/805-4939/6j4m0vn99/index.html>

[S24] Sun/Oracle F77: assumed-size arrays

Array storage sequences and dummy declarations.

<https://docs.oracle.com/cd/E19957-01/805-4939/z40007367227/index.html>

[S25] GNU Fortran: argument passing conventions

ABI conventions, hidden arguments and temporaries.

<https://gcc.gnu.org/onlinedocs/gfortran/Argument-passing-conventions.html>

[S26] LLVM Flang: aliasing in Fortran

The reference-passing intuition does not waive aliasing restrictions.

<https://flang.llvm.org/docs/Aliasing.html>

[S27] Sun/Oracle F77: statement functions

Expression-local procedures and scope.

<https://docs.oracle.com/cd/E19957-01/805-4939/6j4m0vnb9/index.html>

[S28] GNU Fortran: variable FORMAT expressions

Documents the unsupported extension and a replacement.

<https://gcc.gnu.org/onlinedocs/gfortran/Variable-FORMAT-expressions.html>

Primary-source register / 4

Source markers identify documentation or prior work. E01-E03 identify original experiments in the accompanying archive. Proposed architectural consequences and candidate scores are this report's inferences. Vendor manuals describe extensions alongside standard forms; their titles are not conformance guarantees.

[S29] GNU Fortran: ENCODE and DECODE

Not implemented; character internal files are the replacement.

<https://gcc.gnu.org/onlinedocs/gfortran/ENCODE-and-DECODE-statements.html>

[S30] GNU Fortran: extended I/O specifiers

DEC carriage-control compatibility is opt-in.

https://gcc.gnu.org/onlinedocs/gfortran/Extended-I_002fO-specifiers.html

[S31] GNU ld: OVERLAY description

Load/run addresses and the need for an overlay manager.

<https://sourceware.org/binutils/docs/ld/Overlay-Description.html>

[S32] Sun/Oracle F77: list-directed I/O

Null fields, repeated values and slash termination.

<https://docs.oracle.com/cd/E19957-01/805-4939/6j4m0vnc5/index.html>

[S33] GNU Fortran: Cray pointers

Nonstandard address/pointee pairs and aliasing limitations.

<https://gcc.gnu.org/onlinedocs/gfortran/Cray-pointers.html>

[S34] GNU Fortran: TRANSFER

Explicit bit-pattern reinterpretation in Fortran 90 and later.

<https://gcc.gnu.org/onlinedocs/gfortran/TRANSFER.html>

[S35] Bratley and Millo, Computer Recreations: Self-Reproducing Programs (1972)

Software: Practice and Experience 2, 397-400; FORTRAN card-image reproduction is prior art.

https://fab.cba.mit.edu/classes/MAS.865/topics/self_replication/Bratley.pdf

[S36] Nicholas Carlini, IOCCC 2020 printf machine

Author explanation of computation in formatted output. No first-format-computation claim is made here.

<https://www.ioccc.org/2020/carlini/index.html>

[S37] GNU Fortran: evaluation of logical expressions

Evaluation may omit calls; no portable short-circuit side-effect machine.

<https://gcc.gnu.org/onlinedocs/gfortran/Evaluation-of-logical-expressions.html>

[S38] IBM 1401 Fortran Specifications and Operating Procedures, April 1965

H-conversion, printed page 17: input replaces literal characters used by subsequent output; primary manual in an archival collection.

https://ibm-1401.info/C24-1455-2_Fortran_Specifications_and_Operating_Procedures_Apr65.pdf

What GitHub added overnight

The repository now contains a draft research branch, research/carbon-ii-20261008, pinned at 5db3e835d2d14ceef414ae5910a565a08a05bbc. Its CARBON II research was produced after the original exploration. Main still contained only the license when checked. The branch is substantial forward work, not evidence that the original architecture should be discarded. [S41]

A FORMAT description becomes a state selector

```
READ(TABLE,STATE) ROW
READ(ROW,SYMBOL) STATE
```

TABLE contains a finite transition relation assembled before execution. STATE is a 32-character FORMAT description selecting a row. SYMBOL is another FORMAT description selecting a successor description within that row. Reading the successor replaces STATE, so it determines the next row selection. Fortran's native parser and positional A editing perform both selections.

The host runs a finite DO clock, transports input and logs state. Python assembles the records and checks expected traces outside the Fortran transition path. There is no state arithmetic, conditional dispatch or state-indexed application array access inside that path. Selecting a field from a record is still a lookup; assigning it to FORMAT does not make selection disappear.

Artifact	Distinct causal mechanism
CARBON baseline	Numeric word bits become a decimal digit that rewrites a positioning instruction. One integer object is native FORMAT, data, feedback and checkpoint.
CARBON II	Character FORMAT descriptions select rows and successor programs from records. This standard CHARACTER route does not require the baseline's numeric storage bridges.
Shared substrate	Native formatting, record position and mutable executable descriptions. Each experiment keeps its host clock explicit.

What the new capability buys

The supplied examples recognize whether an a-b pattern has appeared and compute cumulative parity of raw binary digits. The pattern input is pre-encoded as selector descriptions; the separate parity specimen handles raw digits. The new suite also demonstrates list-exhaustion control, bounded reversion, indirect addresses, syntax construction, shifts and recurrent composition of individually idempotent maps.

Boundary: the table is 8,192 characters, the row 512, and the program cell 32. The assembler permits at most 16 symbols and states x symbols x 32 <= 8,192. These are finite-state machines with bounded spatial memory and an explicit host clock. No growing tape, general autonomous predicate instruction, unbounded stack or universality is established.

The evidence chain is now complete

Reproduced, rather than accepted on reputation

Measurement	08 October result
CARBON II finite tables	528 traces at each of -00, -02 and -03; 1,584 passed. Corpus hashes match the published run.
External controls	Five representative automata per optimization level passed. The full 528-trace corpus remains internal.
Demos	Internal and external pattern demos passed; raw parity 10110100111 produced 11011000101.
Recovered baseline	37/37 checks passed with static GNU 13.3; another 37/37 passed with shared libgfortran 14.2.
Historical payloads	frames.txt, spool.dat, trace.txt and internal-a.txt exactly match the original bytes under both baseline runtime bindings.

The recovered baseline retains the original 72-line source, tests, specimen and evidence without edits. A wrapper copies it to a temporary directory before rerunning checks, so a new observation cannot silently overwrite the original measurements. The 07 October date embedded in the old harness is a historical report label; the wrapper records the new observation timestamp, compiler command and runtime linkage.

Provenance correction: the original report inferred the runtime from an installed 14.2 package. The recovered compiler setup has no unversioned shared-library link and falls back to its static 13.3 archive. The new shared-14.2 experiment explicitly supplies that link and records the actual core's loaded-library path and hash. Historical baseline documentation is retained as evidence; this update corrects its runtime label.

The comma is compiler physics, not a numeric-storage requirement

For ordinary CHARACTER input containing AB,CDXYZ, -std=legacy internal A8 reads produced AB followed by spaces, while external A8 preserved the full text. With -std=gnu and -std=f95, both media preserved it. The comma-containing internal transducer failed in legacy mode; the colon vocabulary passed. Both outcomes remain in evidence.

GNU 13.3 transfer.c selects a comma-scanning internal-input path when the legacy compile options disable standard warnings. That path does not consult the character-read comma flag. read.c clears that flag for A editing; the external transfer path checks it. The exact source blobs match CARBON II's audit. This explains the CHARACTER control and supports the causal account of the inherited seam; it is not proof about every GNU release. [S39-S40]

The alternative colon representation is valid because the colon descriptor has no terminating effect while an I/O-list item remains. It changes the machine's written vocabulary while preserving the demonstrated finite-table behavior. The result shows one shared construction across media; it does not establish expressive equivalence of all internal and external record machines.

Recovered baseline and next experiments

The forward work remains intact

The reconciliation follow-up builds on the existing CARBON II research branch. It adds baseline/carbon/, tools/reverify_baseline.py, RECONCILIATION.md and separately dated evidence. Its branch is research/carbon-reconciled-20261008; the existing draft PR #1 remains separate. The historical CARBON II results.json and source audit remain unchanged. The original brief's feature archaeology, candidate ranking and architecture stay in this report, with the new findings explicitly updating the frontier.

Three useful next experiments

Stable writable record memory: demonstrate a FORMAT-selected address that can be read, overwritten and read again without host transition logic. Then investigate capacity growth separately. List REWIND, REC=, allocation and record creation as host services, rather than calling them edit descriptors.

Another implementation: test a second compiler/runtime and preserve the same source, records and flags wherever possible. Separate parse acceptance, descriptor behavior, numeric storage extensions and binary spool compatibility. A second GNU runtime version is not an independent compiler implementation.

Input representation: investigate compact raw-symbol selectors and comma-driven rewriting. Compare full states and retain failures. The present pattern assembler supplies executable selectors; the parity specimen demonstrates one raw-digit route, not a general raw-input compiler.

Primary additions to the source register

[S39] GNU 13.3 libgfortran/io/transfer.c

Inspected blob ebad096eeabf7f914751f27711efbee93db9dda6. Compare read_sf_internal and read_sf; the latter checks sf_read_comma.

<https://github.com/gcc-mirror/gcc/blob/releases/gcc-13.3.0/libgfortran/io/transfer.c>

[S40] GNU 13.3 libgfortran/io/read.c

Inspected blob bf2500fc5d050b392144fecbc257ec649dc73b8. read_a disables comma separation during character reads.

<https://github.com/gcc-mirror/gcc/blob/releases/gcc-13.3.0/libgfortran/io/read.c>

[S41] CARBON II source and evidence

Private source repository. Draft PR #1; specimens, instruction_set.csv, assessment.md and evidence/results.json. Independently rerun in this workspace on 08 October.

<https://github.com/0xF1FA/0xF77/tree/5db3e835d2d14ceeef414ae5910a565a08a05bbc>

The result is still a provocation engine. The next contribution should add one capability while exposing its native mechanism. The recovered numeric plate and the newer character selectors give outsiders two distinct doors into the same machine room.